Arjun Narang Final Project Modeling Report

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Objective

The goal of this modeling section is to predict the response variable (Resp) using individual demographic and employment-related characteristics. Preliminary exploratory analysis revealed:

- Nonlinear patterns between Resp and both age and log(wage)
- Bi-modal and right-skewed distribution, violating strict linear model assumptions
- Meaningful interaction between marital status and both age and log(wage)

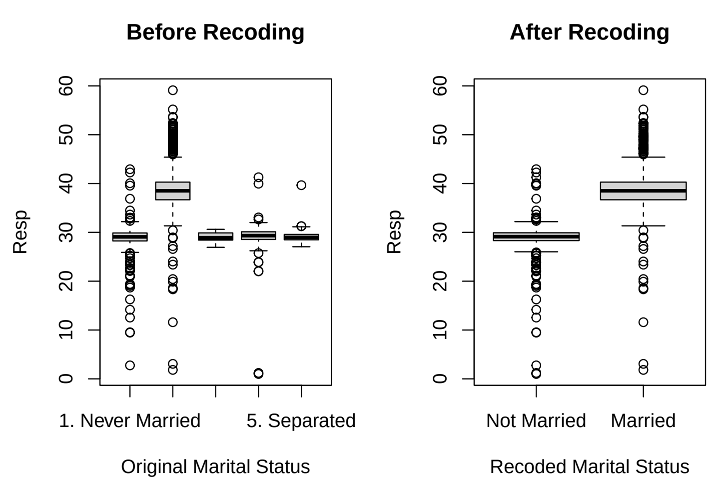
These findings motivated the use of nonlinear regression approaches including polynomial regression, generalized additive models (GAM), and tree-based ensemble methods.

Data Cleaning & Preparation

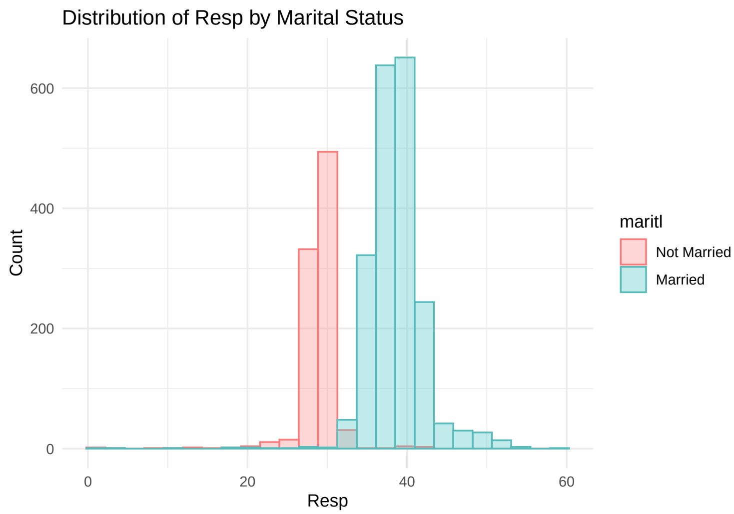
Removing N/A Values:

We see 60 N/A values for Resp in the 3000 values of data. We will remove them as they were missing completely at random.

Recoding Marital Status:



Marital status values were aggregated into Married vs. Not Married to reflect observed differences in Resp.



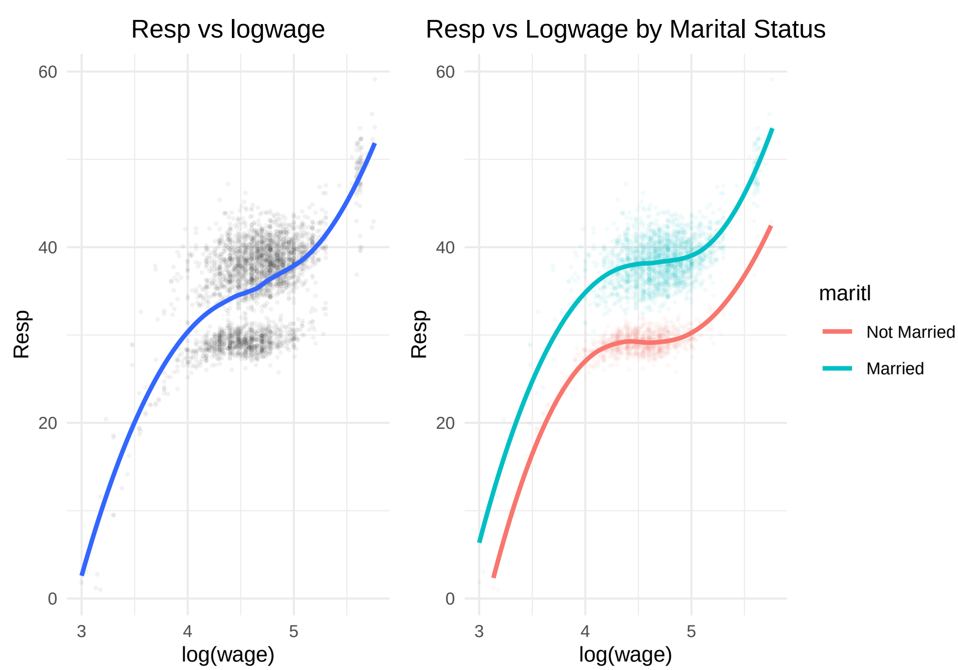
This distinction in marital status also helps define the bimodal distribution observed earlier.

Train/Test Split:

To prepare for the modeling phase, we create an 80/20 train/test split.

Understanding Interactions

Resp vs log(Wage)



While the model without marital status shows one smooth upward trend, the inclusion of marital status produces two distinct trajectories, especially at higher wage levels, illustrating a meaningful interaction effect.

Resp vs Age

Including marital status reveals an interaction effect, where married individuals display consistently higher and more rapidly increasing response values across age.

Modeling Strategy

To evaluate predictive performance and capture the nonlinear and subgroup-dependent patterns observed in the exploratory analysis, multiple modeling approaches were fit in increasing order of flexibility.

Polynomial Regression:

Polynomial regression was first used as a simple and interpretable way to introduce curvature in the model.

- Captures curvature without fully abandoning linear structure
- Used as a first benchmark for nonlinearity
- Fit with and without marital-status interaction terms
- Helps determine whether interaction effects meaningfully improve model accuracy

Generalized Addative Models (GAM):

GAMs were implemented to address the limitations of polynomial models by learning smooth, data-driven functions rather than relying on fixed polynomial shapes.

- Allows for smooth functions rather than assuming polynomial shape
- Supports different smooths by marital status, aligning with observed interaction effects
- Provides interpretability while modeling nonlinear patterns more accurately
- Iteratively expanded to include additional categorical predictors (education, race, jobclass)

Tree Based Methods:

Tree-based approaches were considered to compare parametric models against nonparametric methods capable of automatically detecting interaction structures.

- CART, bagging, random forests, and boosting were tested
- Automatically capture nonlinearities and interaction structures
- Useful as a performance benchmark against GAM
- Less interpretable; performance gain must justify complexity

Model Evaluation Critera:

Model evaluation focused on predictive performance and stability across validation folds.

- Primary metric: Test Mean Squared Error (MSE) on held-out 20% test sample
- Secondary: 10-fold cross-validation for GAM to assess stability
- Considered interpretability when comparing final candidate models
- Model selection balances accuracy, complexity, and real-world explanation

Polynomial Regression Results

Polynomial regression models were first fit to establish a baseline nonlinear relationship before combining them to search for the lowest test MSE.

Table 1: log(Wage) and Age Polynomial Test MSE Results

Model	Test_MSE
Age Polynomial (Degree 2)	23.124524
log(Wage) Polynomial (Degree 3)	19.075852
Age Polynomial (Degree 2) with Marriage Interaction	8.068780
log(Wage) Polynomial (Degree 3) with Marriage Interaction	4.043070
Age and log(wage) Polynomial (no interaction)	14.669246
Age and log(wage) with marital status interaction	1.035746

The results show that adding nonlinear terms for both predictors does not improve performance by much unless interaction effects are included, indicating that subgroup differences by marital status drive much of the predictive gain.

Introducing Categorical Predictors

In an effort to improve the model's performance, the age and log(Wage) model with interaction effects is set as the baseline. From there, various categorical values were tested to gauge an improvement in Test MSE.

Table 2: Top 5 Polynomial Models by Test MSE

Model	Test_MSE
Base + education + race + jobclass + health + health_ins	0.976
Base + education + jobclass + health + health_ins	0.977
Base + education + race + jobclass + health_ins	0.977
Base + education + jobclass + health_ins	0.978
Base + education + race + jobclass + health	0.979

The top-performing polynomial models all include the full or near-full set of categorical predictors, suggesting that demographic and job-related characteristics contribute additional explanatory value beyond nonlinear age and wage effects.

Generalized Addative Models (GAM) Results:

A Generalized Additive Model (GAM) was then fit to replicate the baseline structure used in the polynomial models, incorporating log(Wage) and Age along with their interactions with marital status. Unlike polynomial regression, GAM allows the use of smoothing spline functions, enabling a flexible, non-parametric fit that can better capture nonlinear patterns without imposing a fixed global shape.

Table 3: Top 5 GAM Models by Test MSE

Model	Test_MSE
Base + education + race + jobclass + health + health_ins	0.986
Base + education + jobclass + health + health_ins	0.987
Base + education + race + jobclass + health_ins	0.987
Base + education + jobclass + health_ins	0.988
Base + education + race + jobclass + health	0.989

Similar to polynomial regression, these results suggest that the GAM framework benefits from a full or nearly full list of additional categorical predictors, which enhance predictive accuracy when paired with spline-based nonlinear effects.

Comparison to Polynomial Regression

Compared to the polynomial regression models, the GAM produced slightly higher Test MSE, likely because its smoother spline-based functions constrain flexibility and introduce bias by not attempting to capture sharp curvature in the data. This slight increase in bias does result in a slightly lower residual variance, as observed in the table below.

Table 4: Comparison of Best Polynomial and GAM Models

Metric	Best.Polynomial	Best.GAM
Test MSE	0.976	0.986
10-fold CV MSE	1.031	1.056
Residual Variance	1.013	1.010
Total Effective DF	22.000	31.110

The best polynomial model achieves slightly lower test MSE (0.976 vs 0.986) and lower 10-fold CV MSE (1.031 vs 1.056) than the GAM, indicating marginally better out-of-sample performance and stability with fewer effective degrees of freedom (22 vs 31.11). Although the GAM attains a very slightly lower residual variance, this comes at the cost of greater complexity, so the polynomial model offers a better balance of accuracy, parsimony, and interpretability.

Tree Based Method Results

Tree-based approaches were implemented to compare the performance of parametric models against non-parametric methods capable of automatically detecting nonlinearities and interaction structures. CART, bagging, random forests, and boosting were fitted to the data, allowing for flexible modeling of the response variable without the need to specify polynomial terms or interactions explicitly.

Table 5: Comparison of Tree Based Methods

Model	Test MSE	CV MSE	Residual Variance	Effective DF
CART (Pruned)	3.528	4.253	3.428	8.000
Bagging	1.472	1.596	0.319	2233.342
Random Forest	1.604	1.512	0.363	2239.581
Boosting	1.419	1.641	1.190	9834.000

Looking at the table above, we can see that Boosting has the lowest Test MSE, but not the lowest CV MSE, residual variance, or effective degrees of freedom.

- CV MSE: Random Forest has the lowest (1.512), slightly better than Boosting (1.641).
- Residual Variance: Bagging and Random Forest have much lower values (0.319 and 0.363), indicating they fit the training data better than Boosting (1.190).
- Effective Degrees of Freedom: CART (Pruned) is by far the simplest, with only 8, while Boosting is the most complex (9834).

So, Boosting is best for predicting on new data (lowest Test MSE), but it is more complex and has higher training error than Bagging and Random Forest. The low residual variance and CV MSE for Bagging and Random Forest suggest they may be more stable and less prone to overfitting, but they don't generalize quite as well to the test set as Boosting in this case. CART, although by far the simplest, performed the worst among the four tree-based methods.

Pros of Tree Based methods:

Tree ensembles automatically capture nonlinear and complex interactions without requiring explicit specification of polynomial terms or interaction effects (like we have done for the GAMs and polynomial regression models), which is appealing because of the interaction between marital status, age, and log(wage) observed in the exploratory analysis. They also handle mixed numeric and categorical predictors naturally and give variable-importance summaries that can highlight which demographic and job-related features drive predictions.

Cons of Tree Based methods:

However, despite this flexibility, none of the tree-based models are able to reach the very low test MSE achieved by the best polynomial model, nor the slightly higher but still competitive GAM, meaning they offer no predictive advantage here. In addition, the effective degrees of freedom for ensembles such as boosting are many orders of magnitude larger than those of the polynomial and GAM models, indicating a much more complex fit with little to show in terms of improved accuracy.

Interpretability and stability:

The parametric models are also preferred due to their interpretability: the polynomial model derives a concise equation with a small number of coefficients, whereas boosted trees, for example, need to aggregate thousands of small trees whose combined effect is extremely difficult to explain qualitatively. Moreover, the GAM and polynomial fits exhibit stable behavior under cross-validation with relatively low CV MSE and residual variance, while the tree ensembles introduce additional tuning complexity (number of trees, depth, learning rate) without giving any clearly better bias–variance trade-offs.

Include Table Here for 4 Methods

- Test MSE
- CV MSE
- Residual Variance
- Effective Degrees of Freedom

Short summary of best one.

Comparison to Parametric Models

Explain pros but mostly cons because it doesn't perform as well

Model Evaluation

- Explain how close the polynomial and GAM are in results (compare bias and variance). Lowkey this is basically the section I did above (Summary Comparison of Best Polynomial and GAM Models) so maybe it would be better here – perhaps a copy paste scenario. You'd have to add the tree results too, just pick the best of the tree methods (boosting I think)
- Looks like Polynomial has the edge in Test MSE and CV MSE. Although it barely loses the variance battle, its effective DF is so much lower than the GAM spline method.
- Talk about explainability of polynomial method vs others.
- Write out final Model Equation

Appendix Stuff